

Business Requirement Document (BRD)

Credit Risk & Loan Default Analysis Dashboard

1. Project Overview

Financial institutions must evaluate loan applications accurately to minimize credit risk while maintaining profitable portfolio growth. Existing manual or rule-based assessments often fail to capture hidden behavioral and historical risk patterns. To improve decision-making, the organization requires a centralized analytics solution that consolidates customer demographics, loan history, credit exposure, and repayment behavior into a unified, data-driven view.

This project aims to design and develop an **end-to-end analytics solution** that processes raw financial data and delivers actionable insights through SQL-based analysis and an interactive Power BI dashboard.

2. Business Objectives

The key objectives of this project are:

- Identify **high-risk and low-risk customer segments**
- Measure **overall portfolio health and default rate**
- Analyze the impact of **customer demographics, income, and employment** on credit risk
- Evaluate how **past loan history and payment behavior** affect default probability
- Enable stakeholders to make **data-driven lending decisions**
- Improve **risk monitoring and reporting efficiency**

3. Stakeholders:

Stakeholder	Role
Risk Analysts	Assess borrower risk
Credit Managers	Approve/reject loans
Business Analysts	Generate insights & reports
Management	Strategic decision-making
Data Team	Data processing & dashboard development

4. Scope

In Scope

- Data cleaning and transformation using SQL & Power Query
- Aggregation of transactional data to the customer level
- Customer risk feature engineering
- SQL-based business analysis
- Power BI dashboard creation
- Interactive visualizations and KPIs

Out of Scope

- Machine learning modeling
- Real-time scoring system
- Production deployment
- External data integration

5. Data Sources

The solution uses the following datasets:

Table	Description
customer_application	Customer demographics & loan status
previous_application	Historical loan applications
bureau	External credit exposure
installments	Loan repayment behavior
credit_card_balance	Credit card utilization history

All transactional tables are aggregated to the **customer level (SK_ID_CURR)** for analysis

6. Functional Requirements

The system must:

Data Preparation

- Clean missing and inconsistent values
- Standardize data formats
- Create derived metrics (age, utilization, late payments)
- Aggregate transactional tables to customer level

Analysis

- Calculate default rates and KPIs
- Perform customer segmentation
- Identify behavioral risk indicators

- Support SQL-based ad-hoc queries

Dashboard

- Display portfolio overview metrics
- Show demographic risk analysis
- Provide credit behavior insights
- Enable filtering via slicers (income, age, education, etc.)
- Provide drill-down capability

7. Key Business Questions

The dashboard should answer:

1. What is the overall loan default rate?
2. Which customer segments are most risky?
3. Does income or education impact repayment behavior?
4. Do previous refusals increase default probability?
5. How do late payments affect credit risk?
6. Does high credit card utilization correlate with defaults?
7. Which customers represent low risk with high loan potential?

8. Non-Functional Requirements

Requirement	Description
Performance	Dashboard loads within 5 seconds
Scalability	Handle 300k+ customers efficiently
Accuracy	Aggregated metrics must match source data
Usability	Easy-to-understand visuals
Maintainability	Reusable queries and transformations

9. Success Metrics

Project success will be measured by:

- Accurate calculation of default metrics
- Clear identification of risk segments
- Fast dashboard performance
- Actionable insights for stakeholders
- Reduced manual reporting effort

10. Tools & Technologies

- SQL Server – Data querying & analysis
- Power BI – Visualization & dashboard
- Power Query – Data transformation
- Excel – Validation & testing

11. Expected Deliverables

- Cleaned and structured dataset
- Aggregated customer-level tables

- SQL analysis queries
- Power BI interactive dashboard
- Project documentation

12. Conclusion

This solution provides a structured, scalable, and business-focused approach to credit risk assessment by integrating multiple financial data sources into a unified analytical framework. The dashboard empowers decision-makers with real-time insights, enabling smarter lending strategies and improved portfolio performance.